
EDUCATION

University of Hawai'i at Manoa, Honolulu, HI (3.82 GPA)**Aug 2022 – Dec 2026**

B.S. Mechanical Engineering

Mountain View High School, Mountain View, CA**Aug 2018 – June 2022**

WORK EXPERIENCE

Hawai'i Institute of Geophysics & Planetology: Materials Research Assistant**October 2024 – Present**

- Designed and manufactured fixtures for precision polishing of diamond anvil cells and machining of ceramic heating elements.
- Prepared Diamond Anvil Cell samples inside an inert atmosphere glovebox.
- Configured a CNC lab automation system based on the science Jubilee platform to perform sonication experiments; used SolidWorks for design and SLA 3D printing for most manufacturing.

Makai Ocean Engineering: Engineering Intern**October 2023 – May 2024**

- Designed multiple components for subsea micropile drill rig including pressure compensator system, adjustable laser guide, and micropile tapping jig.
- Worked to fabricate & assemble a variety of prototype equipment on multiple projects.
- Assisted in terrestrial systems tests for micropile drill rig.

Cyclehop LLC: Bike/Scooter Mechanic**June 2022 – August 2022**

- Managed and repaired a fleet of ~50 electric bikes.
- Worked independently to organize and maintain a small warehouse.

SELECTED PROJECTS

Formula SAE - Suspension Lead (Senior Design)**August 2025 – Present**

- Led a team of students and volunteers to design a suspension system for a racing vehicle.
- Programmed vehicle dynamics and tire models in Python to optimize vehicle performance in steady state cornering and acceleration.
- Created robust, equation driven assemblies of the suspension in SolidWorks and communicated with other team members to integrate them with the chassis and drivetrain.
- Used Python to perform structural calculations on control arms.
- Fabricated a variety of components including control arms, steering column, differential mounts, chassis sections, etc. using shop tools, welders, lathes, and mills.

Stair Climbing Robot (Sophomore Design)**January 2024 – March 2024**

- Worked with a 5 person team to design and build a robot to climb two flights of stairs, retrieve an egg, and return the egg to the bottom of the stairs.
- Responsible for architecting the full-system CAD model in OnShape and manufacturing laser cut and 3D printed components.
- Performed structural analysis and optimization on sandwich panel wheel design with a suspension system which allowed the robot to handle high impact loads.

Neural Network**March 2022**

- Wrote a multi-layer neural network to recognize hand-written characters using Java.
- Utilized multithreading to write algorithms to handle vector operations and back propagation.
- Tested and compared different activation functions to improve speed and performance of the network.

SKILLS

CAD Modeling : SolidWorks, Fusion, OnShape, & Free CAD**Prototyping/Fabrication:** 3D printing, tig/mig/stick welding, laser cutter, waterjet, metal fabrication, some machining**Programming:** Proficient in Python and MATLAB, additional experience in Java, Rust, C#, Arduino, & C; Familiar with Data Structures & OOP Principles**Miscellaneous:** Technical Communications, Microsoft Word & Excel